11.3 The Real Pendulum

WORKSHEET

The question

STEP III 1998, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q11

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The bob moves on a circle of radius l. What are its speed and its height (relative to the lowest point) in terms of θ and $\dot{\theta}$?

STAGE 1 Energy

Covers: the energy equation

Ask yourself Write the speed and the height of the bob in terms of θ and $\dot{\theta}$. The bob is at rest when $\theta = \gamma$. Use conservation of energy.

STAGE 2 The period as an integral

Covers: the period

Ask yourself Use $\cos\alpha = 1 - 2\sin^2\frac{\alpha}{2}$ on both cosines. What fraction of the period is spent going from $\theta = 0$ to $\theta = \gamma$? Write the time as $\int \frac{d\theta}{\dot{\theta}}$.

STAGE 3 Removing the singularity

Covers: the small-amplitude approximation

Pause and try Put $k = \sin\frac{\gamma}{2}$ and $\sin\frac{\theta}{2} = k\sin\phi$. What range of ϕ do you need? Find $d\theta$ in terms of $d\phi$. Which factors cancel?

STAGE 4 The first correction

Covers: the small-amplitude approximation

Ask yourself For small k, expand $(1-k^2\sin^2\phi)^{-1/2}$ by the binomial theorem and keep two terms. Then use $\int_0^{\pi/2}\sin^2\phi\,\mathrm{d}\phi$, and $k \approx \frac{\gamma}{2}$.

What's really going on

Elliptic integrals

The integral

$$K(k) = \int_0^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}}$$

is the *complete elliptic integral of the first kind*, so called because a close relative gives the perimeter of an ellipse. The exact period is $P = 4\sqrt{l/g}\,K(\sin\frac{\gamma}{2})$. Expanding the integrand fully and integrating term by term using the Wallis integrals $\int_0^{\pi/2}\sin^{2n}\phi\,\mathrm{d}\phi$ of Section 4.4, one finds

$$P = 2\pi\sqrt{\frac{l}{g}}\Big(1 + (\tfrac{1}{2})^2k^2 + (\tfrac{1\cdot3}{2\cdot4})^2k^4 + (\tfrac{1\cdot3\cdot5}{2\cdot4\cdot6})^2k^6 + \cdots\Big).$$

No finite combination of familiar functions gives $K(k)$; it is a genuinely new function.

Gauss's remarkable formula

There is a far better way to compute K. In Section 2.6 you met the *arithmetic–geometric mean*: start with a and b, and repeatedly replace them by their arithmetic and geometric means. Gauss discovered in 1799 that

$$K(k) = \frac{\pi}{2\,\mathrm{AGM}(1, \sqrt{1-k^2})},$$

so the period of a pendulum with amplitude γ is exactly

$$P = \frac{2\pi\sqrt{l/g}}{\mathrm{AGM}(1, \cos\frac{\gamma}{2})}.$$

The AGM converges so fast that, for swings of up to about 160°, three or four steps give the period to ten significant figures, and the values in Figure 11.3.1 (in the full section online) were computed this way. Nearer 180° the two starting values, 1 and $\cos\frac{\gamma}{2}$, are far apart and it takes a step or two more, but five steps suffice even at 179°. Gauss wrote in his diary that this discovery "opens an entirely new field of analysis", and he was right: inverting elliptic integrals led Abel and Jacobi to *elliptic functions*, and those led to *elliptic curves*, which are now central to number theory (they were the key to the proof of Fermat's Last Theorem) and to the cryptography that secures the internet.

Galileo's lamp and Huygens's clock

Galileo's claim that a pendulum's period does not depend on its amplitude, made around 1602, was a good approximation for the small swings he studied. Christiaan Huygens realised that it was not exact, and in 1659 found the curve along which the period really is independent of amplitude: the cycloid, the subject of Section 11.4. His pendulum clocks, made from 1656 onwards, improved timekeeping from errors of about a quarter of an hour a day to about fifteen seconds. Marine chronometers, which solved the problem of finding longitude at sea, later abandoned the pendulum altogether, but pendulum clocks remained the most accurate timekeepers in the world until the 1920s.

Try this next

1 **A tethered particle.** Try STEP III 2015, Question 9 (step.maths.org/questions/15-s3-q9): an oscillator whose period depends on its amplitude even for small oscillations. Why does it, when a pendulum's does not?

2 **The next term.** Carry the binomial expansion one term further and show that $P \approx 2\pi\sqrt{\frac{l}{g}}(1+\frac{\gamma^2}{16}+\frac{11\gamma^4}{3072})$.

3 **The AGM by hand.** Use three steps of the AGM to estimate the period of a pendulum with amplitude 90°, and compare with the value 1.1803 in the figure.

4 **Clocks.** A pendulum clock is regulated to keep perfect time with an amplitude of 4°. If its amplitude increases to 5°, how many seconds a day does it lose?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 4, which treats the large-amplitude pendulum.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). Gauss's discovery and its uses, including the fastest ways to compute π.
- **Dava Sobel, *Longitude*** (Fourth Estate). The story of timekeeping at sea, beyond the pendulum.

The full guided solution is in Section 11.3:
https://beyondstep.keithmcnulty.org/ch11/real-pendulum.html